

Concept Plans and Spatial Organization Diagrams

How the park is organised — the crescent, the radial alleys, and the rooms between them.

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Concept Development

Al Safa 2 Park — Alternatives, Evaluation & Final Concept Selection

AI-GENERATED DRAFT — FOR REVIEW. Three distinct concept alternatives are developed and scored, each responding differently to Phase 3's objectives.

4.1 Research Inspiration

- **Local context:** Dubai's Shamal wind pattern, palm/Ghaf shade traditions, sikka (shaded alleyway) urban form.
- **International precedents (typology reference, not copied):** shaded desert park strategies using canopy structures and misting (Middle East precedent), linear park zoning with clear activity bands (High Line-style legibility), biophilic play design for inclusive playgrounds.
- **Landscape trends:** climate-responsive/native xeriscaping, multi-generational activity zoning, park-as-microclimate-infrastructure thinking.

4.2 Mood & Design Language Direction

Material and form language: warm sand/stone tones, filtered dappled shade (not solid roofed structures everywhere), local Ghaf/palm planting as the dominant canopy species, and a restrained geometric language (straight + arced paths) echoing the site's elongated rectangular geometry rather than fighting it.

4.4 Concept Alternatives

	Concept A — "Shaded Spine"	Concept B — "Canopy Village"	Concept C — "Cool Loop"
Organizing idea	One continuous shaded central spine connects all zones	Cluster of discrete shaded "rooms" (play, fitness, social, quiet) around a plaza	A single perimeter shaded loop (jogging/walking) with activities nested inside
Circulation	Linear, highly legible, single primary path	Radial from central plaza to each room	Circular loop + inner cross-paths
Shade strategy	Continuous overhead shade structure along the spine	Shade concentrated per-room (tree clusters + pergolas)	Shade concentrated on the loop itself
Best fits	Wayfinding clarity, day/night activation along one axis	Distinct age-group zoning, event flexibility	Fitness/wellness emphasis, continuous exercise use
Risk	Areas off the spine may still be under-shaded	Central plaza could become a pinch-point / less flexible for events	Interior zones could feel disconnected from the loop's comfort

4.5 AI-Assisted Exploration

All three concepts were tested against Phase 1.06's shadow-length data (6m canopy shadow ranges from ~0.2m at summer noon to ~19m at winter evening) to sanity-check shade-structure spacing assumptions before scoring. This is documented further in Phase 9 (AI Workflow).

4.6 Evaluation Matrix

Criterion (weight)	A: Shaded Spine	B: Canopy Village	C: Cool Loop
Function (25%)	8	7	8

Criterion (weight)	A: Shaded Spine	B: Canopy Village	C: Cool Loop
User Experience (25%)	8	9	7
Sustainability (20%)	7	8	7
Feasibility within AED 35M (20%)	9	6	8
Innovation (10%)	7	8	6
Weighted Total /10	7.85	7.55	7.35

4.7 Final Concept Selection

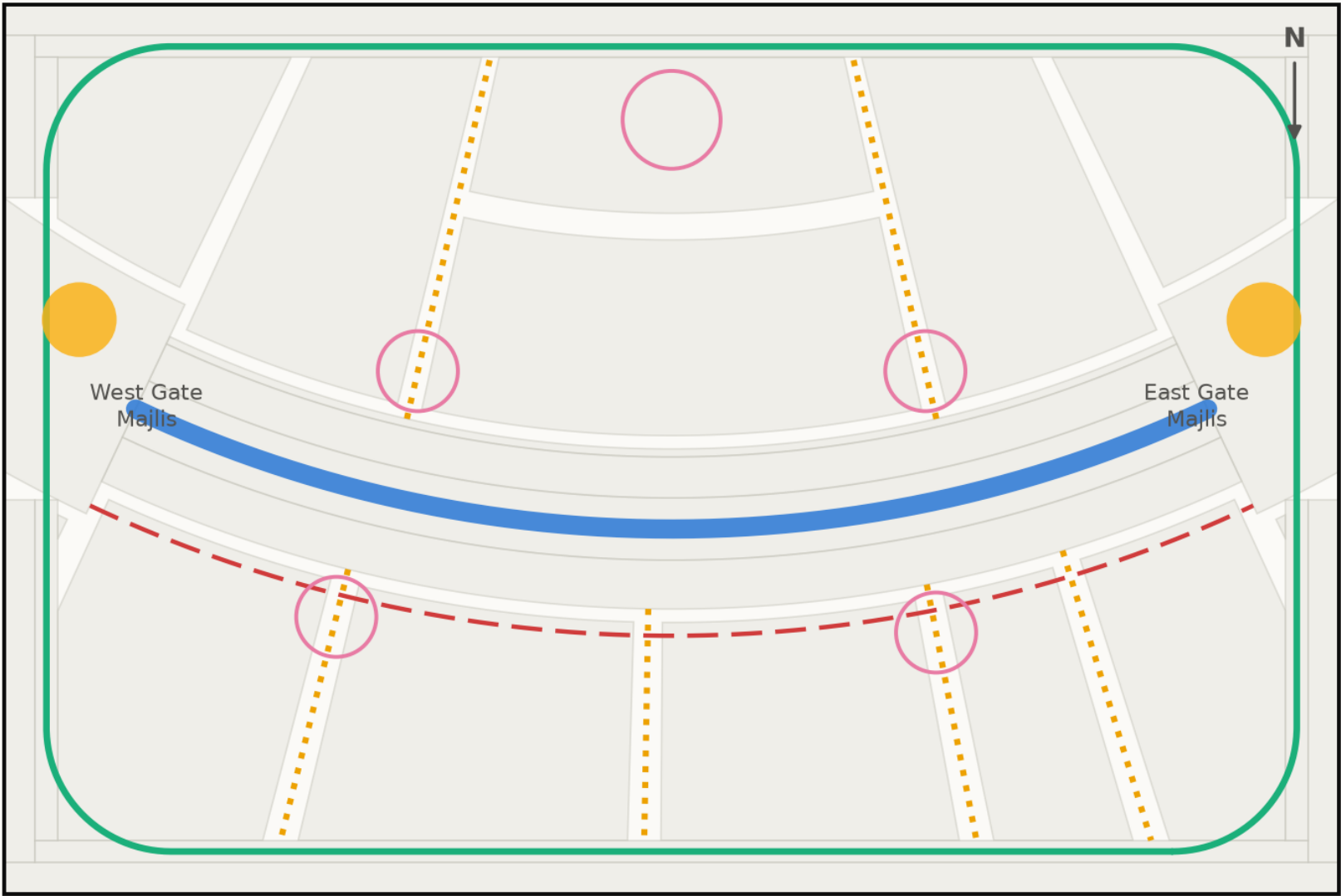
Selected: Concept A — "Shaded Spine", with Concept B's room-based zoning logic merged in as secondary structure along the spine (best of both: legibility + distinct age-group zones). Justification: highest feasibility score within the fixed AED 35M budget, directly addresses Phase 2's #1 evidence-backed problem (summer thermal comfort) via one continuous engineered shade structure rather than dozens of smaller ones, and gives the AI Design Challenge submission the clearest single diagram to communicate (important per Evaluation Matrix criterion 6: Quality of Presentation, 5% weight, and criterion 4: Quality of Design and User Experience, 15%).

Circulation

Technical drawing — to scale.

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free
- Al Sikkak — shaded alleys to the rooms
- Al Madar — 438 m running loop
- Service / emergency vehicle access
- Majlis — shaded rest, never more than 33 m from the walk

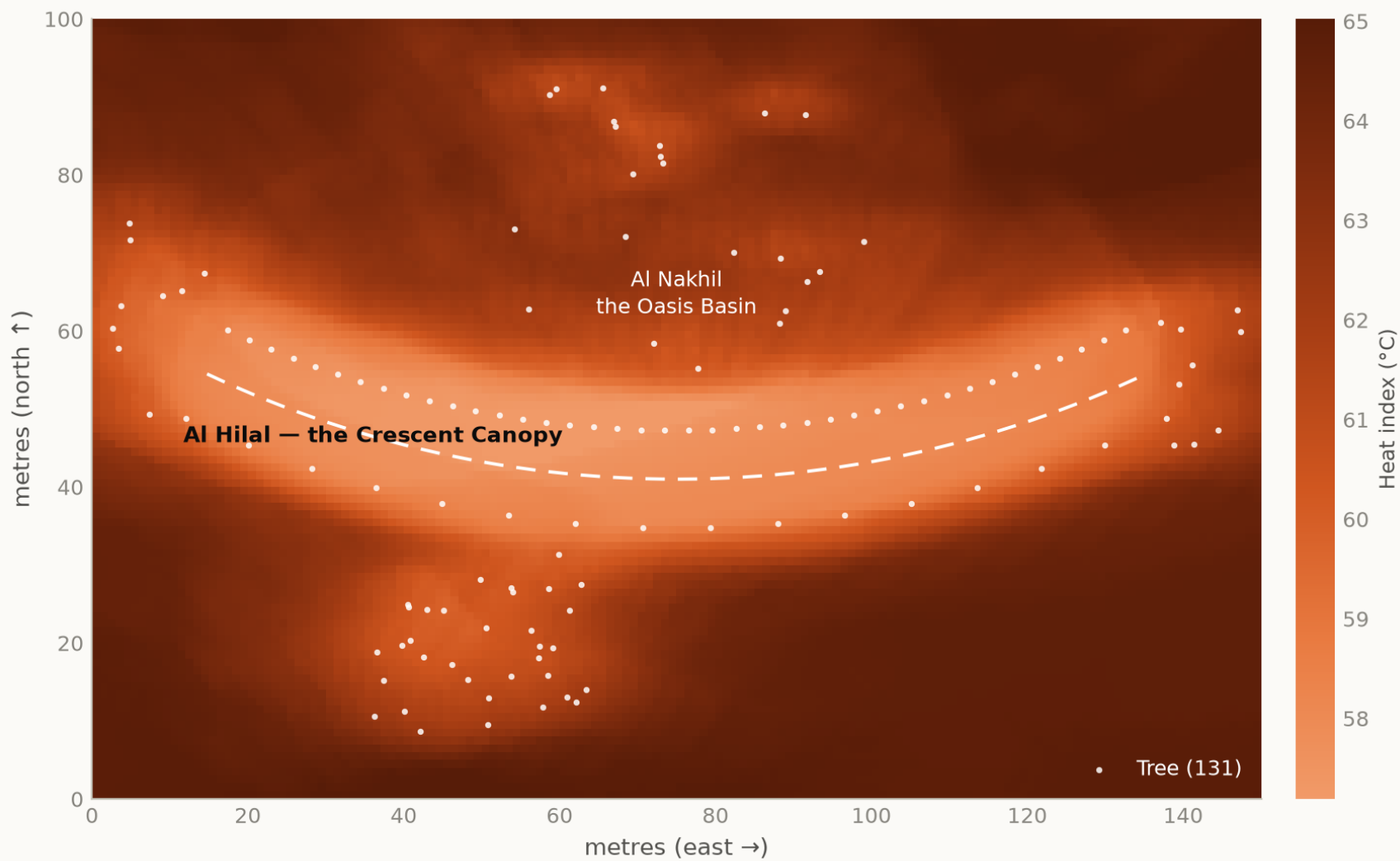
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig04 site comfort map

Analysis output — computed from project data.

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



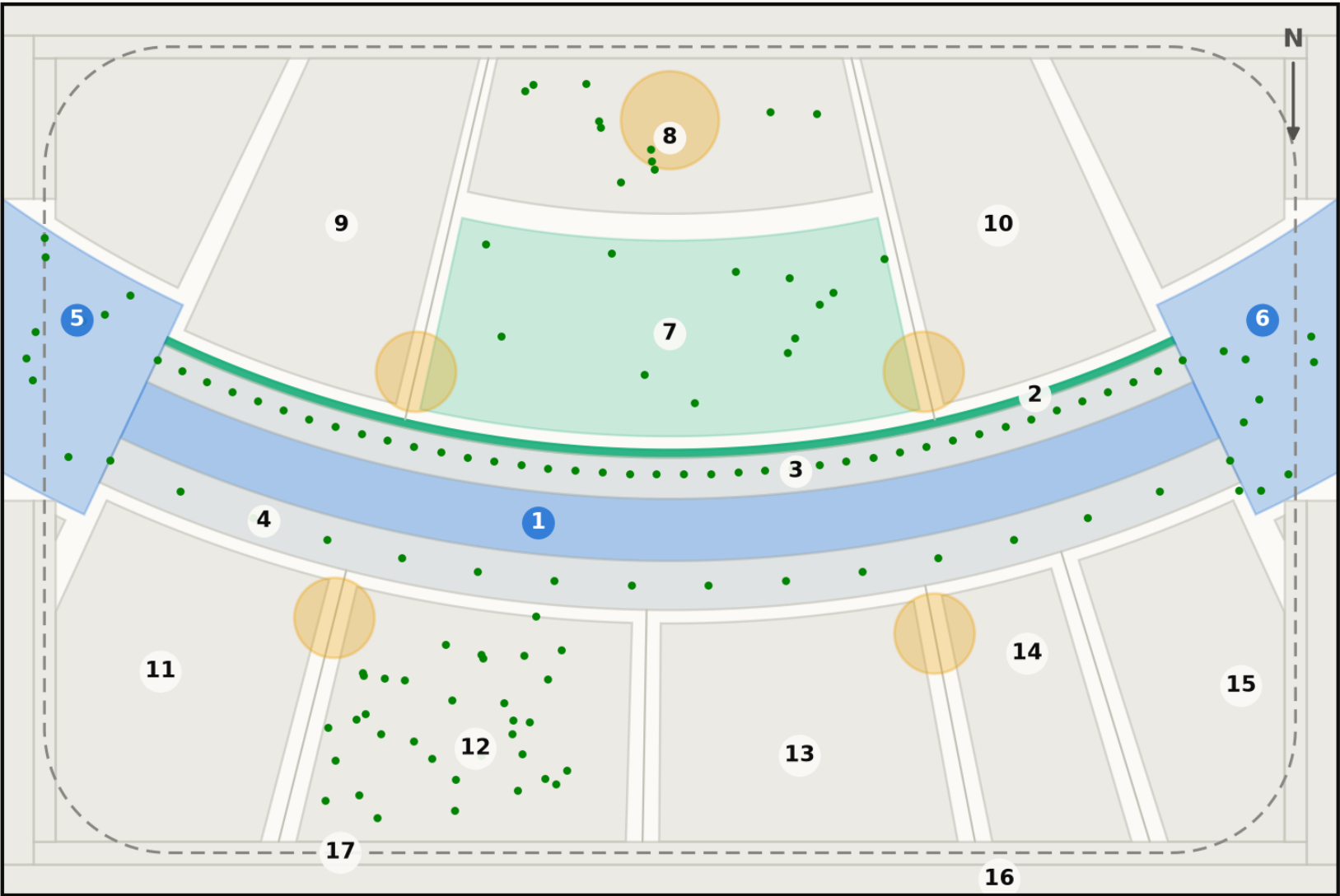
Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.
Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
Al Sikkak — shaded alleys & setbacks		1,414 m²

Total 15,000 m²

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge